

# ARSHIN SIKKA

Singapore | +65 8016 4894 | arshin.sikka@u.nus.edu | linkedin.com/in/arshin-sikka | github.com/arshinsikka

## EDUCATION

**National University of Singapore**

**Aug 2023 - May 2027**

**Bachelor of Computing in Computer Science**

- Artificial Intelligence Focus Area - Certificate of Distinction
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Computer Organisation, Linear Algebra, Probability & Statistics, Computer Vision, NLP, AI Planning & Decision Making, Intro to AI & ML
- Teaching Assistant, AI Ethics and Intro to AI & ML — Designed and facilitated weekly tutorials and labs for ~50 undergraduates across two courses

## SKILLS

- Programming Languages: Python, Java, TypeScript, JavaScript, SQL, HTML/CSS
- Backend & Databases: FastAPI, Node.js, Spring Boot, REST APIs, Socket.IO, PostgreSQL, Supabase, Prisma
- Infrastructure: Docker, Fly.io, CI/CD, Git/GitHub, Bitbucket
- Testing & Security: pytest, JUnit, JWT authentication, RBAC, multi-tenant isolation, VAPT remediation
- Frontend: React, Next.js, Angular, Tailwind CSS
- AI Systems: LangChain, LangGraph, Langfuse, RAG pipelines, MCP, multi-agent orchestration, LLM evaluation, guardrails
- ML & Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Monte Carlo simulation

## WORK EXPERIENCE

**AI Engineer Intern, Garuda Robotics, Singapore**

**Jul 2026 - Present**

- Architecting an agentic mission-command system end-to-end — conversational client, MCP server, and governance gateway — that turns plain-language requests into validated drone commands, blocking unverified model output from reaching flight hardware.
- Designing a multimodal video intelligence pipeline that pairs computer vision with LLM summarisation, surfacing evidence clips and mission summaries from live drone footage.

**Data Science Intern, SP Digital (Data & AI), Singapore**

**Jan 2026 - Jun 2026**

- Engineered a context-aware guardrail layer for a production LLM assistant serving 100+ grid fieldworkers, closing third-party penetration test findings spanning prompt injection, persona-bypass jailbreaks, and SQL schema disclosure.
- Built an automated adversarial evaluation pipeline using Langfuse covering 400+ attack prompts across multiple regression datasets, cutting unsafe model responses by ~60% across internal deployments.

**AI Engineer Intern, KPMG, Gurugram, India**

**May 2025 - Aug 2025**

- Designed an agentic RAG system using LangChain and Azure OpenAI embeddings over a FAISS vector store, replacing manual SharePoint search across hundreds of consulting documents with conversational retrieval.
- Augmented the retrieval engine with custom document-parsing layers to cite source PDFs and extract structured Excel data, cutting consultant knowledge lookup time 40-50% across a 12-person team.

**SDE Intern, AlygnAI, California, USA (Remote)**

**Jun 2025 - Aug 2025**

- Migrated an early-stage AI product from a no-code Bubble prototype to a production FastAPI backend, implementing JWT authentication, refresh tokens, bcrypt hashing, and 2FA.
- Benchmarked LoRA against RAG for LLM adaptation, influencing the founding team's core product architecture decisions.

**SWE Intern, StatusNeo, Gurugram, India**

**May 2024 - Jul 2024**

- Developed secure REST APIs in Java Spring Boot for a large Indian bank's digital transformation, implementing JWT authentication and RBAC with JUnit test coverage.

## PROJECTS

**Socratic Digital Twin, NUS Faculty of Dentistry**

**Jun 2026 - Present**

- Architecting the backend for an AI tutor that trains orthodontics residents through Socratic questioning, encoding senior specialists' reasoning strategies into a 10-node LangGraph pipeline over a pgvector retrieval layer.
- Building a model-agnostic gateway routing between OpenAI, Claude, and DeepSeek through provider adapters, with a guardrail layer that regenerates any response breaking the questioning-only constraint.

**Singapore Society Simulation, NUS Odyssey ([GitHub](#))****May 2026 - Aug 2026**

- Developed a multi-agent policy simulation platform grounded in Nvidia Nemotron Singapore personas, supervised by Prof See-Kiong Ng and selected for public academic showcase at NUS School of Computing.
- Engineered multi-round deliberation with belief drift tracking, where agents reason from their prior position and an LLM-generated summary of opposing views, benchmarked against real Reddit discourse on the same policy questions.

**Lecture AI, Co-Founder, Singapore ([Website](#))****Jul 2025 - Mar 2026**

- Shipped an end-to-end AI pipeline converting lecture recordings into structured study notes, chaining audio preprocessing, Whisper transcription, and a two-pass correction and summarisation layer at under \$1 per lecture.

---

**LEADERSHIP**

---

**Director of Human Resources, NUS Student Union (NUSSU)****Nov 2024 - Present**

- Designed onboarding and feedback systems for NUS's apex student body, supporting 100+ student leaders across cross-functional teams.

**Professional Player & Team Captain, Chess****Apr 2015 - Present**

- Represented India at the 2019 Commonwealth Chess Championship and placed runner-up in a FIDE-rated tournament of 693 players; captained NUS to Inter-Faculty Games victory.